

# Summer 25 MATH 4100 Exam 1 Study Guide

The exam will cover lectures 8-28 to 9-22 and will be held during class on Monday 9-29. If you have testing accommodations, you will receive an email with further instructions.

**Exam Makeup:** Questions will consist of multiple choice, true/false, computational and proof type questions. Partial credit will be given as appropriate. The exam will be closed note and you will be expected to memorize all relevant results.

## List of Topics:

- Vector spaces and subspaces
- Sums and direct sums of subspaces
- Inner products
- Linear independence, Span, and Bases
- Orthogonal complements
- Linear maps
  - range and kernel
  - Bijectivity and isomorphism
  - Matrix representation
- Matrices
  - Matrix algebra
  - Solving  $A\mathbf{x} = \mathbf{b}$
  - Rank and Nullity
  - Fundamental subspaces
  - Incidence matrix and directed graphs
  - Change of basis
  - Projections

**Suggested Exercises From Master Problem List:** All types of computational problems are fair game. Here are a list of proof/conceptual problems to familiarize yourself with: 4b, 7, 10, 19+20, 23, 30, 33, 43, 44